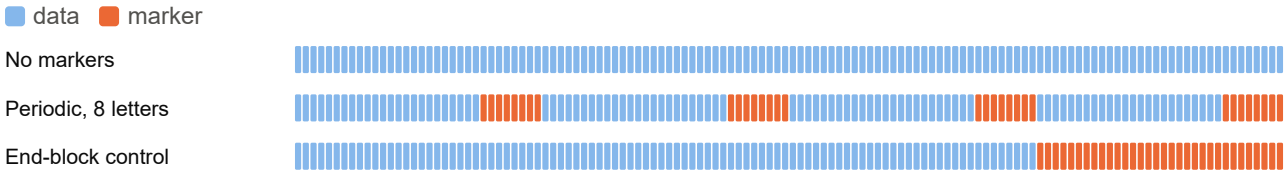
Do synchronization markers help DNAFormer with deletions?

FYP progress report · results of fine-tuning DNAFormer on 8 strand layouts · test set of 1,000 strands per deletion level

Question. In DNA storage, deleted letters shift everything after them, which makes strands hard to recover. If we place short known sequences (**markers**) at fixed positions in each strand, does DNAFormer recover strands better, and is it worth the space the markers use?

What we tested

Every strand is 128 letters. Markers take space away from data, so longer markers leave fewer data letters. All versions use the **same data, the same simulated errors and the same training** (pretrained DNAFormer, fine-tuned for 10 epochs). Only the markers change.



We tested 4 markers of length 4, 5, 6, 7, 8 and 12 letters, one at the end of each 32-letter block, plus no markers. The **end-block control** uses the same 8-letter markers, all placed together at the end: if markers help only because there are fewer data letters to recover, it would score as well as the spread-out version.

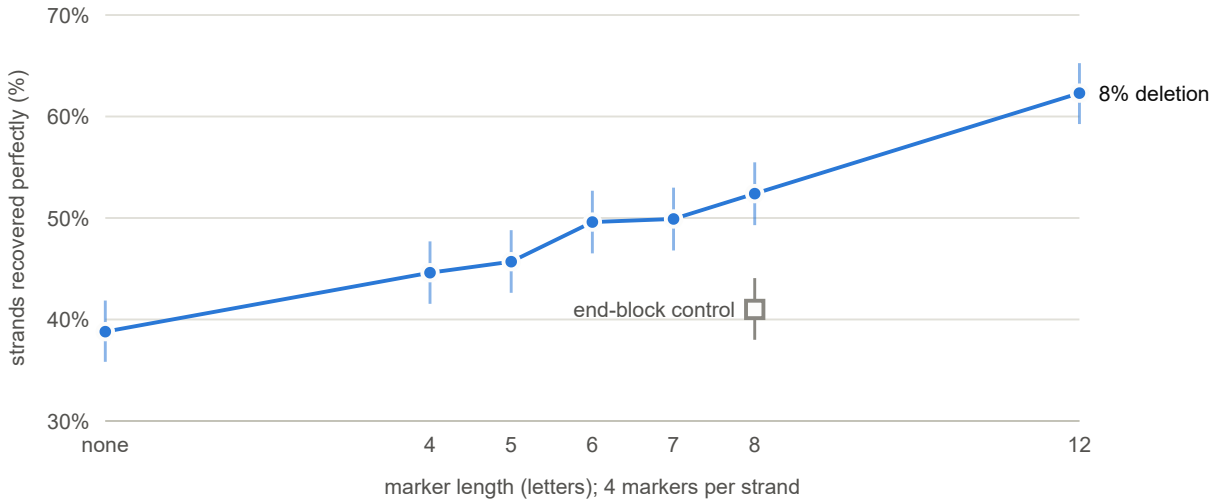
Main results (8% deletion, the hardest case)

38.8% → 62.3%	4.9% → 1.9%	≈ 50
strands recovered perfectly: no markers → 12-letter markers	letters still wrong after decoding	data letters recovered per strand, for **every** marker length

1. **Markers make strands more reliable, and longer markers help more.** Perfect recovery rises from 38.8% (no markers) to 52.4% (8 letters) and 62.3% (12 letters). 12 vs 8 is a clear difference ($p < 0.001$), so we have not yet found where it stops improving.
2. **The benefit comes from where the markers are placed.** Spread out: 52.4%. Same markers bunched at the end: 41.0% ($p < 0.001$). So the markers really help DNAFormer re-align reads after deletions.
3. **But markers do not increase the total data recovered.** Each marker letter costs one data letter, and the reliability gained almost exactly makes up for it: every version recovers about 50 data letters per strand at 8% deletion.
4. **At low deletion (1–3%), markers make no difference**: every version recovers 99.5–100% of strands. At 5%, the version without markers recovers the most data.

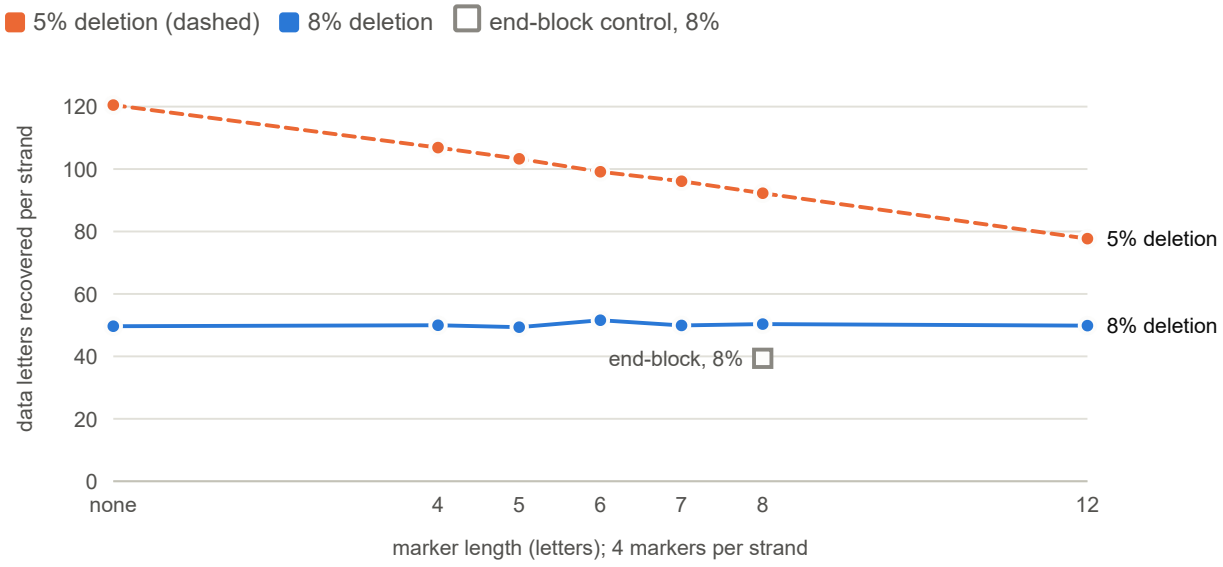
Results in two charts

1. Strands recovered perfectly at 8% deletion



Each dot is one fine-tuned model, tested on 1,000 strands; bars show 95% confidence intervals. The gray square is the end-block control (same markers as length 8, placed at the end).

2. Data letters recovered per strand



Perfect strands × data letters per strand. At 8% the line is flat: what markers gain in reliability, they lose in space.

All results

Marker length	Data letters	Marker share	Perfect at 5%	Perfect at 8%	Wrong letters at 8%	Data recovered at 8%
none	128	0.0%	94.1%	**38.8%**	4.9%	50
4	112	12.5%	95.4%	**44.6%**	3.8%	50
5	108	15.6%	95.6%	**45.7%**	3.7%	49
6	104	18.8%	95.3%	**49.6%**	3.4%	52
7	100	21.9%	96.1%	**49.9%**	3.4%	50
8	96	25.0%	96.1%	**52.4%**	2.6%	50
12	80	37.5%	97.1%	**62.3%**	1.9%	50
8, at end	96	25.0%	95.7%	**41.0%**	5.2%	39

At 1% deletion every version is 100%; at 3%, 99.5–100%. "Data recovered" = perfect strands × data letters per strand.

What this means

> Markers are a **real synchronization aid**: they make each strand much more reliable under heavy deletions, and the effect comes from their placement. However, **on their own they do not store more data**, because the reliability they add matches the space they take. Whether they are worth using depends on the **error-correcting code** used on top. An error-correcting code adds extra letters so that mistakes can be fixed. Markers lower the number of mistakes, so fewer extra letters may be needed. That trade-off is what we want to test next.

Limitations

- Each version was trained once. Small gaps (for example 6 vs 7 vs 8 letters, 1–3 points) may be training noise.
- Data and errors are simulated, with deletion rates (5–8%) much higher than real nanopore reads.
- One marker sequence per length. Because the stored data is random, the exact letters of a marker should matter little; only its length was varied.

Proposed next steps

1. **Markers vs error correction at equal cost:** compare (a) no markers + more error-correction letters with (b) markers + fewer error-correction letters, using the same total extra space.
2. **Find where the benefit stops:** test 16-letter markers.
3. **Confirm the results:** paired statistical tests and a second training run for the best lengths.